



Subject.....AC 2.....Department.....HCT.....

S No	Experiment Name	Date	Marks (32)				Faculty Sign
			Performance (14)	File (10)	Attendance (8)	Total (32)	
1.	To familiarize camp in lab	19/1/26	12	8	8	28	
2	To study op-amp (74) as inverting & non-inverting amp	2/2/26	12	7	8	27	
3	Observe, sketch plot of input wave of op-amp PC differential eqn. det.	16/2/26	12	7	8	27	
4	To study op-amp as active ultra low voltage follower, cal. of voltage gain	23/2/26	12	8	8	28	
5	Construct Port of notch circuit	6/3/26	12	8	8	28	
6	To study & sketch shifting wave of Schmitt	23/3/26	10	7	8	25	23-3-26
7	To study waveform of Schmitt Trigger circuit	6/4/26	10	8	8	26	
8	To make & test observation of Astable Multivibrator using 555 timer	6/4/26	11	8	8	27	
9	Design & analysis of 1st order low pass filter	13/4/26	11	8	8	27	
10	To study 2nd order low pass filter	14/4/26	12	8	8	28	24-4-26

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Expt

- To familiarise the op-amp (IC 741) & equipment used in the analog elec lab and also design open loop op-amp
- Uses, 741 (IC) op-amp, probes, resists. (2.2)k Ω , breadboard, DC supply.

The operational amplifier is a direct coupled, high gain volt. amplifier with very high input impedance and o/p impedance.

The applications of operational amplifiers has become so widely diversified that this original terminology is inappropriate since op-amp are now used as a basic building block for phase shifting, filtering, signal conditioning, multiplying, detecting etc.

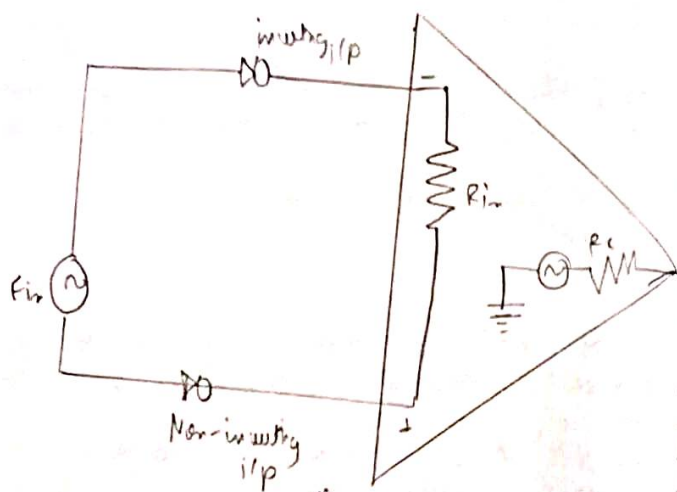
Equivalent ch. for an Ideal Operational Amplifier

- 1) The V gain is infinite $\therefore A_{v0} = \infty$
- 2) The i/p res R is ∞ $\therefore R_{in} = \infty$
- 3) The o/p R is 0 $\therefore R_o = 0$
- 4) The B.W is ∞ $\therefore BW = \infty$
- 5) There is 0 i/p offset volt $\therefore E_o = 0$ if $E_{in} = 0$

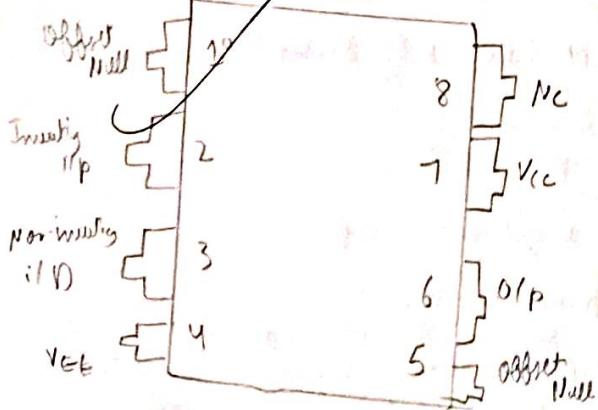
1) Open loop non-inverting
uses an op-amp where the i/p signal goes to the (+ve) terminal, the (-) terminal is grounded (no feedback)
$$A(v) = \frac{V_{out}}{V_{in}}$$

Teacher's Sign.

Fig. 1. Ckt for an ideal op-amp.



Pin Diagram of IC 741C



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1) Open loop inverting

Just an op-amp where the i/p signal goes to the (-ve) terminal & the (+ve) terminal is grounded but w/o any feedback connection from o/p to i/p.

Introduce to Instruments used

1.) Digital Storage Oscilloscope (DSO)

It's an instrument which stores a digital copy of the waveform in the digital memory which it analyses further using digital processing.

2.) Function Generator

It's piece of elec test equipment or software used to generate elec waveform.

3.) Dual Power Supply

The 30V, 2A dual power supply has been designed as constant current (CC) & constant volt (CV).

It provides floating & o/p volt. It is ideally suitable for complex analog & digital applications.

Thus, the IC 741 op-amp and analog electronics laboratory equipment were familiar and the open-loop and op-amp was designed & verified successfully.

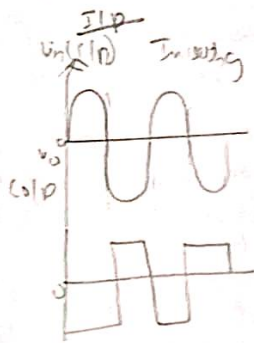
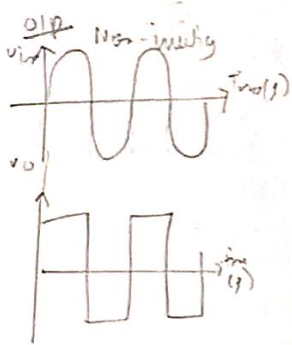
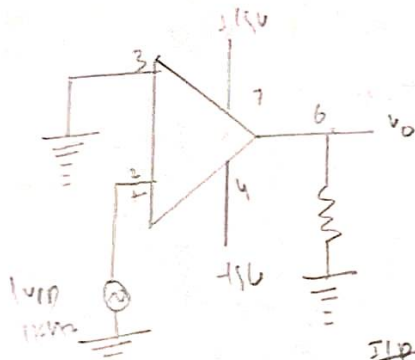
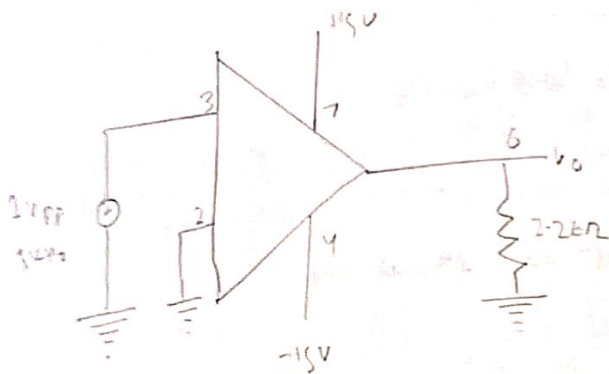
1.) Check the connections carefully.

2.) All connections must be tight.

3.) Off the power supply after completion.

4.) Do it carefully & avoid damage.

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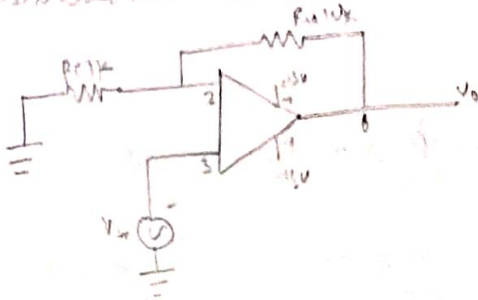
* Table comparing op-amp ideal values & op-amp datasheet values

No.	Parameters	Ideal op-amp value	Datasheet op-amp values
1)	Voltage gain (open loop gain)	Infinite	$\approx 100,000 - 200,000$ V/V
2)	Input impedance	Infinite	$\approx 2\text{M}\Omega$
3)	Output impedance	0	$\approx 75\Omega$
4)	Common mode	Infinite	$\approx 70 - 100\text{dB}$
5)	Slew Rate	Infinite	$\approx 0.5\text{V}/\mu\text{s}$
6)	Offset Voltage	0V	$\approx 1\text{mV}$

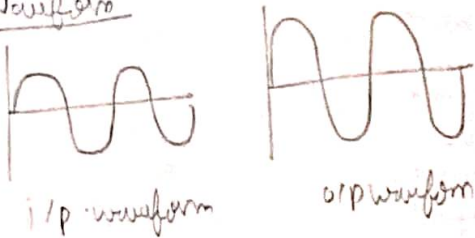
5- The IC 741 operational amplifier was studied and its basic characteristics were understood.

Teacher's Sign.

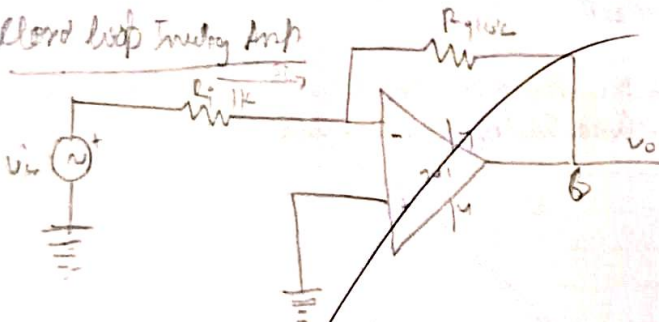
Closed Loop Non-Inverting Op-Amp



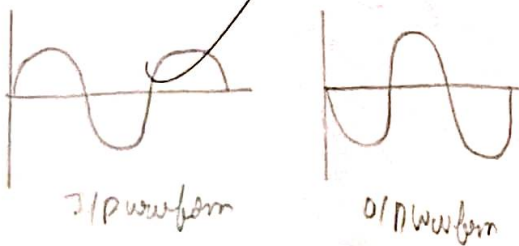
Waveforms



Closed Loop Inverting Amp



Waveforms



phase diff. b/w i/p & o/p = 180°

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Exp-2

- 1- To study the op-amp (IC-741) as inverting and non-inverting amplifier and calculate its gain.
- 2- Due to very large loop gain of op-amp, in open loop configuration, op-amps go to saturation with i/p signals slightly greater than 0.7V. So it is not suitable for linear applications and open loop configurations have limited app.

a) Inverting amplifier configuration

It's a closed loop mode applⁿ of op-amp and employs -ve feedback chain. The R_f and R_i are the feedback and i/p resistors of the op-amp respectively.

The pot difference across the i/p terminals of an op-amp is a negⁿ of the large open loop gain. Due to these 2 reasons, the inverting terminal is at virtual ground i.e. 0.

$$\therefore I_i = I_f$$

$$\text{i.e. } V_{in} / R_i = -V_o / R_f$$

$$\therefore V_o / V_{in} = A_v = -R_f / R_i$$

Here, the -ve sign indicates that the op-amp will be an ant amplifier with 180 degree phase shift.

Hints for Design of inverting amplifier.

Gain of an inverting amplifier

$$A_v = V_o / V_{in} = -R_f / R_i$$

Let the req. gain = 10,

$$\text{i.e. } A_v = R_f / R_i = 10$$

Let $R_i = 1k\Omega$, then $R_f = 10k\Omega$

OBSERVATIONS

Inverting

V_{in}	V_o
500mV	4.66V
1V	9.38V
1.5V	14.56V
2V	19.74V
2.5V	24.92V
3V	27.2V
3.5V	27.2V
4V	27.2V
4.5V	27.2V
5V	27.2V

Practical gain:

$$A_v = \frac{V_o}{V_{in}} = -9.68$$

Theoretical gain

$$A_v = -\frac{R_f}{R_i} = -\frac{10k\Omega}{1k\Omega} = -10$$

OP saturated at 27.2V
offset 0.16V

Non-Inverting

V_{in}	V_o
500mV	5.8V
1V	11.2V
1.5V	17V
2V	22.4V
2.5V	20.8V
3V	27.2V
3.5V	27.2V
4V	27.2V
4.5V	27.2V
5V	27.2

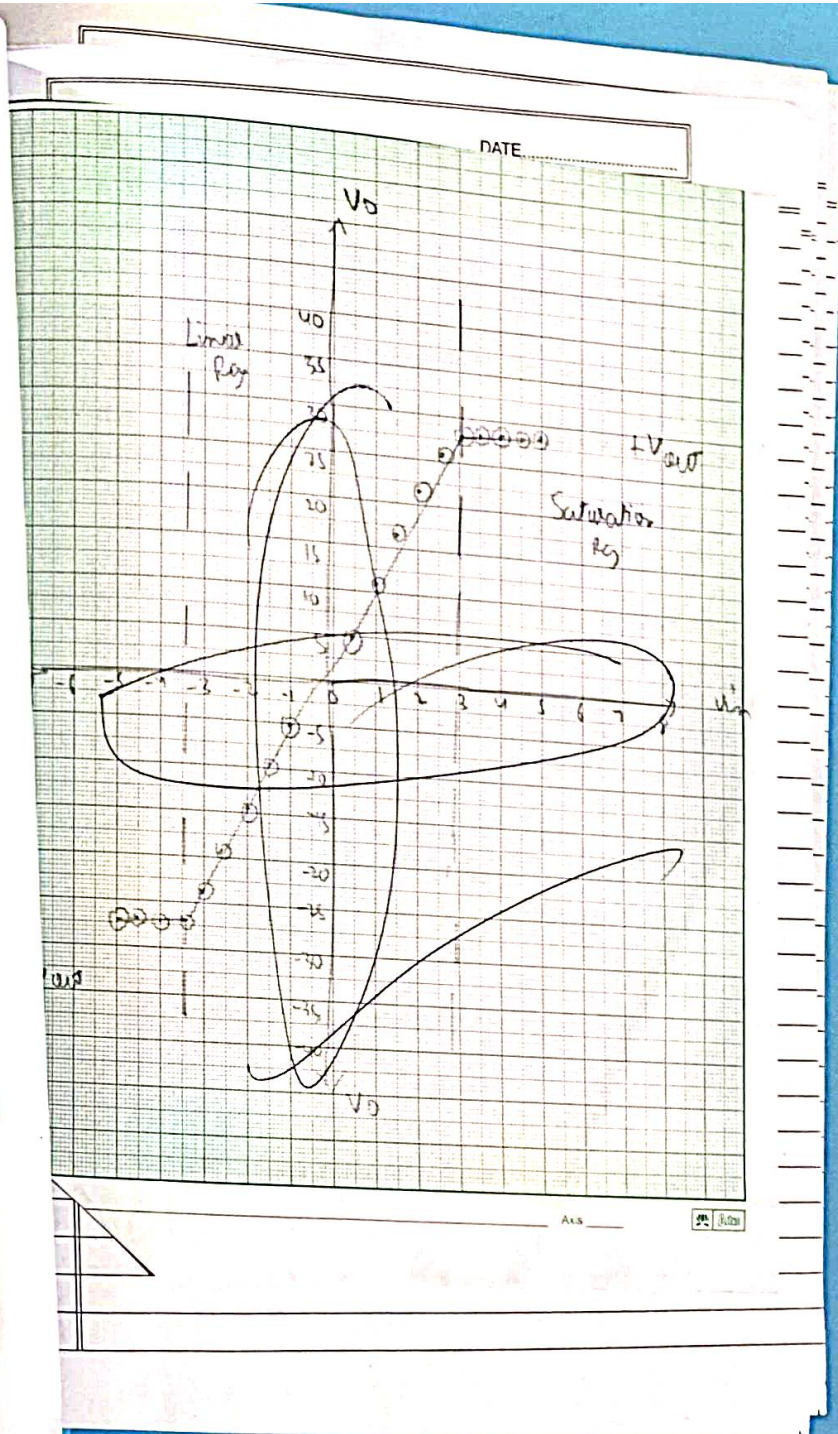
Practical gain:

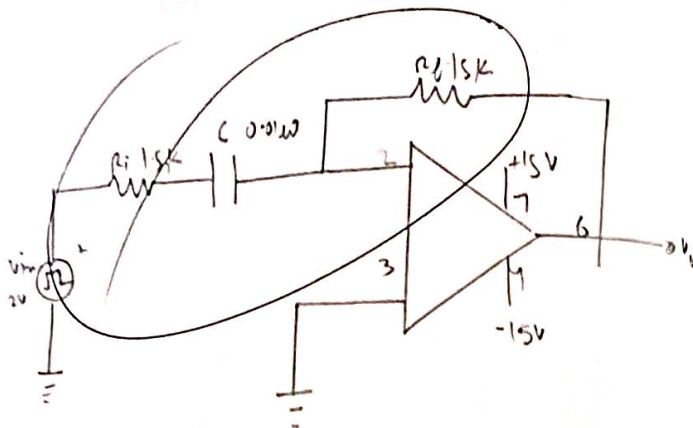
$$A_v = \frac{V_o}{V_{in}} = 11.2$$

Theoretical gain

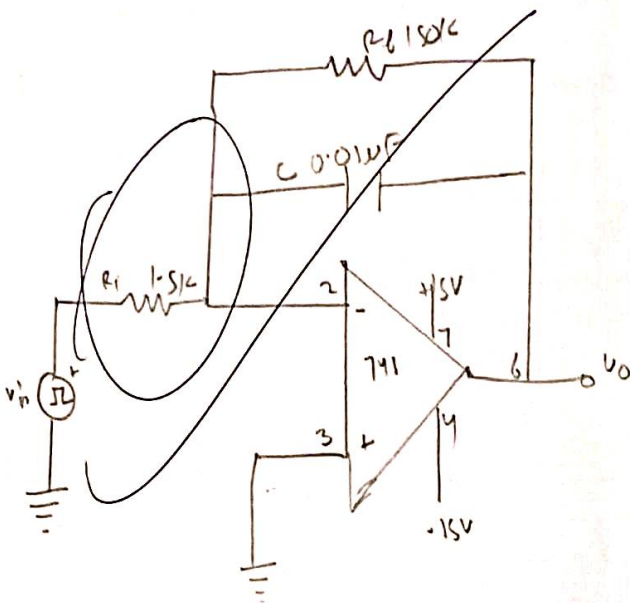
$$A_v = 1 + \frac{R_f}{R_i} = 1 + 10 = 11$$

0° phase difference





Ckt diagram of differentiator



Ckt diagram of integrator

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Exp-3

Observe and plot the output wave shapes of op-amp RC differentiating ckt, RC integrating for square wave input.

Components used

IC [741], breadboard, resistors, $10k\Omega$, $1k\Omega$, $22k\Omega$, capacitor $0.01\mu F$, DC supply

a) Differentiator

It's an op-amp ckt. which performs mathematical operation of derivative or differentiation of the i/p voltage. That is $V_o = -R_f C \frac{d(V_i)}{dt}$.

The ckt also works as high-pass filter

b) Integrator

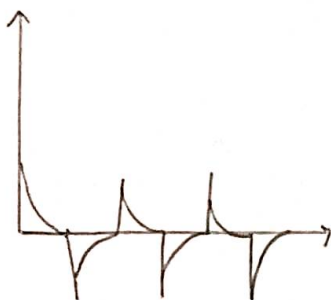
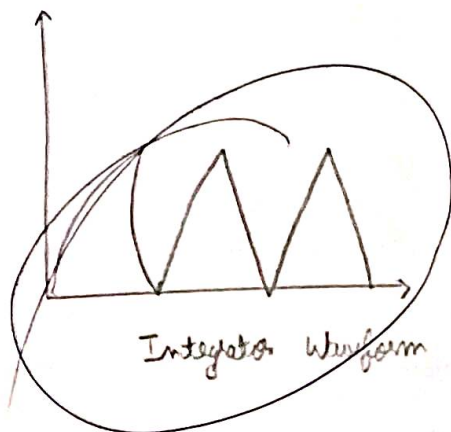
It's an closed loop op-amp ckt which performs the mathematical operation of integration. given by $V_o = (-1/R_f C) \int V_i dt$.

This ckt. also works as low pass filter

c) For square wave input

i) In RC differentiation ckt

The output is a series of +ve & -ve spikes at the rising & falling edges of the square wave.



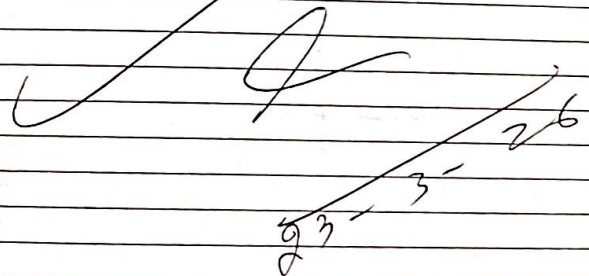
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2) In R-C integrated ckt
In O/p is a series of triangular waveform corresponding to the integration of the square wave input.

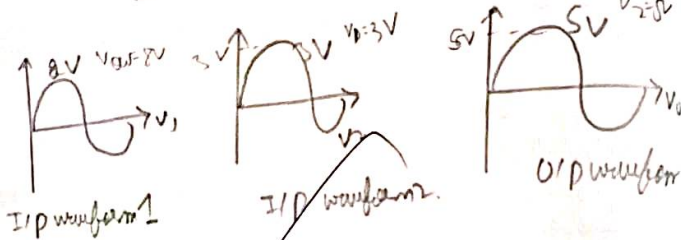
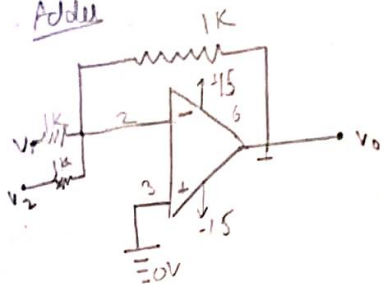
Precautions

- 1) Check all the connections carefully before switching on the supply.
- 2) Avoid loose connections
- 3) IC should be handled carefully to avoid damage.
- 4) Ensure resistors are of proper values as per the ckt.

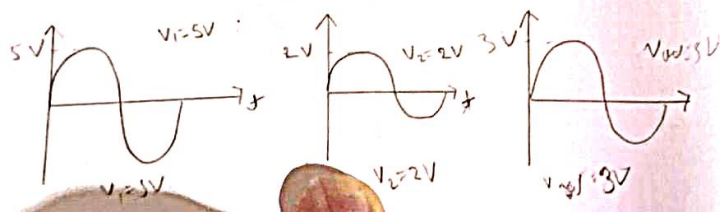
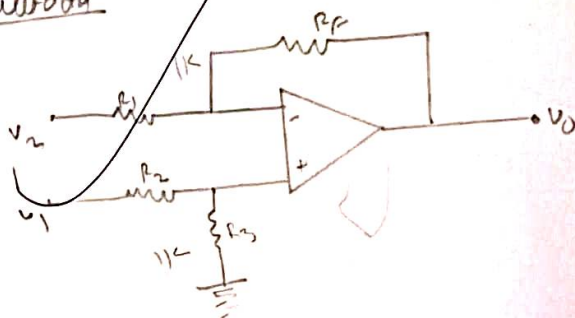
The o/p waveforms of the op-amp RC differentiating & RC integrating ckt's were observed & plotted successfully.



Adder



Subtractor



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Exp-4

Aim: To study the op-amp (IC-741) as adder, subtractor and voltage follower, calculate the o/p voltage.

Comps: IC-741, breadboard, resistors, DC supplies.

Obj: The elec. ckt which perform arithmetic operation called arithmetic ckt. using op-amp we can build some arithmetic ckt's such as an adder and a subtractor.

Def: An adder is an electronic ckt. that produces an o/p which is equal to the sum of applied i/p. According to its virtual concept, the volt. of the inverting i/p terminal of an op-amp is same as that of the volt. at its non-inverting i/p terminal.

The virtual eqⁿ at its inverting i/p terminal

$$(0-V_1)/R_1 + (0-V_2)/R_2 + (0-V_0)/R_F = 0$$

$$\Rightarrow V_1/R_1 - V_2/R_2 = V_0/R_F$$

$$\Rightarrow V_0 = R_F (V_1/R_1 - V_2/R_2)$$

$$\text{if } R_F = R_1 = R_2 = R$$

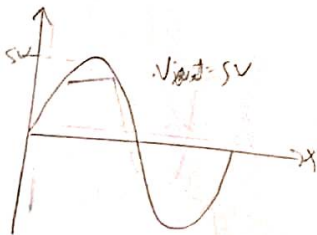
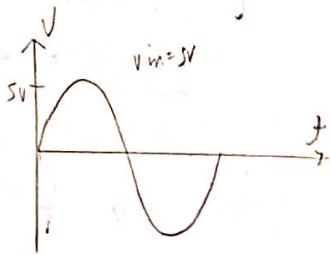
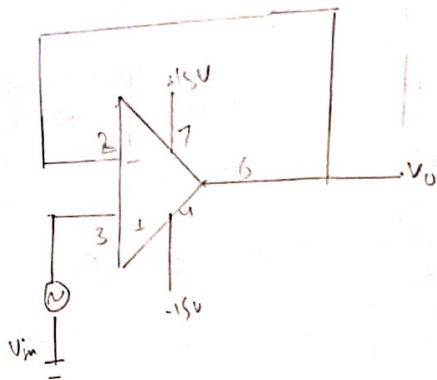
$$\text{Then } V_0 = -R (V_1/R - V_2/R)$$

$$\Rightarrow V_0 = -(V_1 - V_2)$$

Subtractor Ckt:

A subtractor is an electronic ckt. that produces an o/p which is equal to the difference of applied inputs.

Voltage Follower



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This If $R_f = R_1 = R_2 = R_3 = R$, then the voltage follower is:

$$V_o = V_i - V_2$$

Voltage follower

A volt. follower is an op-amp ckt. which has a volt gain of 1. This means op-amp does not provide amplification to the input.

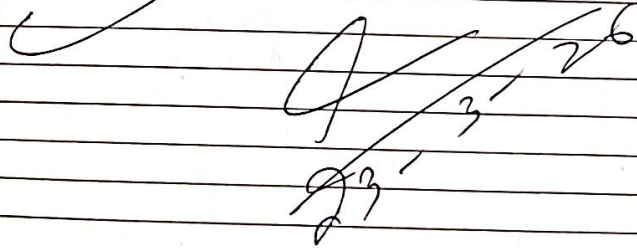
The volt. follower is a non-inverting amplifier with unity gain.

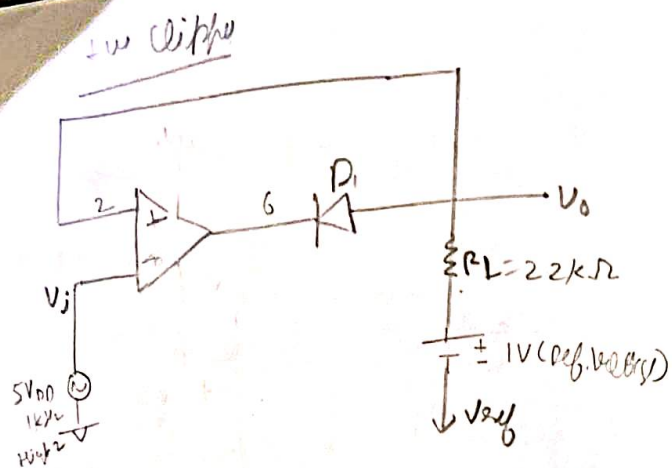
$$A = 1 + R_f/R_1 = 1$$

$$\text{or } R_f/R_1 = 0$$

$$\therefore R_f = 0$$

Let - The o/p of the op-amp and address, subtractor, voltage follower is studied and plotted above.





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Exp - 5

mc- Construct positive and negative peak clipper ckt.

ach- IC-741, breadboard, resistors, Diode, DC supply.

ay- A clipper is a elec. ckt. that produces a copy removing a part of the i/p above or below a reference value. ~~Due to this, the peak to peak amplitude of o/p of clipper will be always less than that of input.~~

Clippers can be classified into the following types based on the clipping portion of the i/p.

i) Positive clipper, ii) Negative clipper

i) Positive clipper

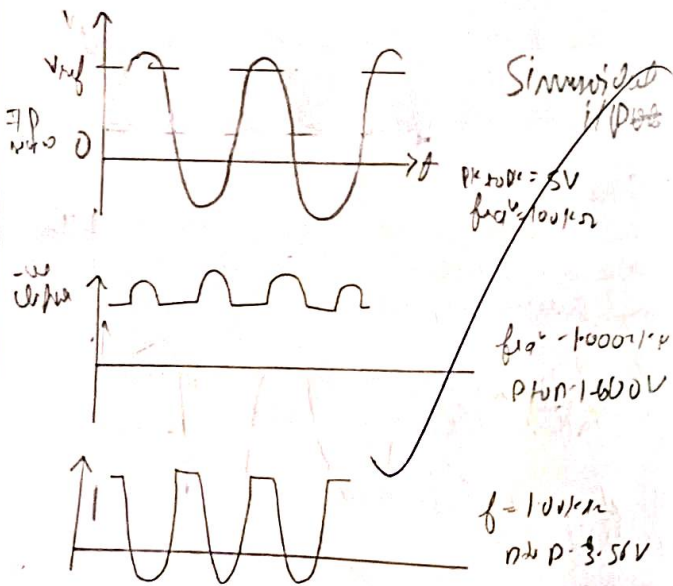
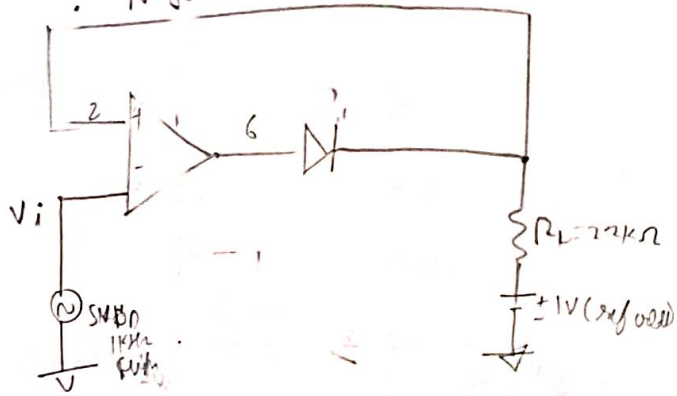
A positive clipper is a ckt. that clips the +ve part of i/p above a reference volt. V_{ref} allows the remaining portion to pass.

$V_i < V_{ref}$, diode conducts & o/p follows i/p ($V_o = V_i$)

$V_i > V_{ref}$, diode is OFF, o/p is limited.

~~This ckt. clips the +ve half of i/p signal at reference volt. level~~

Negative Clipper



• T/P, O/P waveform of -ve clipper.

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2) Negative clipper

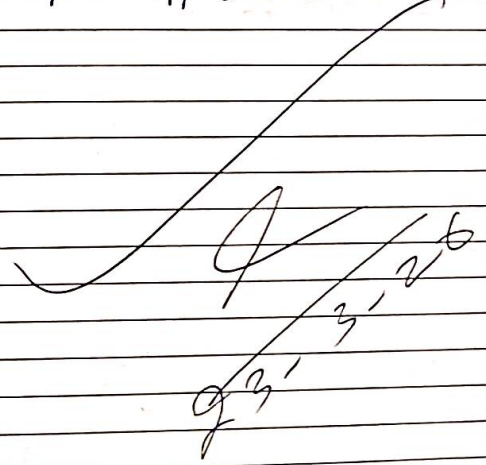
A negative clipper is a circuit that clips the negative portion below reference volt. V_{ref} and allows the positive remaining part to pass.

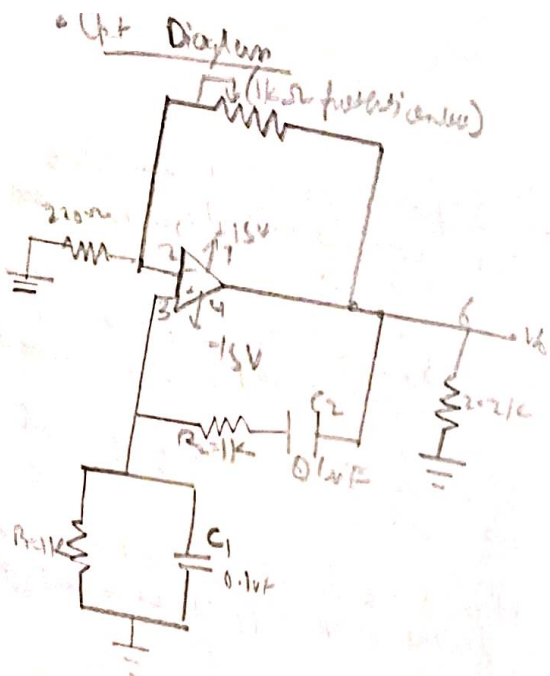
$V_i > V_{ref}$, diode conducts, o/p follows input ($V_o = V_i$)

$V_i < V_{ref}$, diode is off, o/p limited to reference volt V_{ref} .

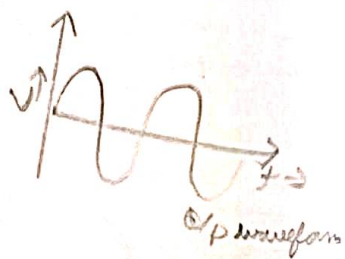
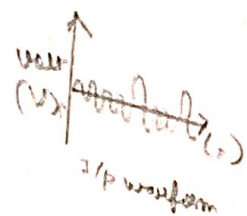
Thus, the circuit clips the negative half of input as reference volt level.

The construction & o/p waveform for the negative peak and positive peak clippers are studied and verified.





Result



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Exp-6

To study RC phase shift Wein Bridge oscillator measurement of freq & amplitude of oscillations using op-amp.

There are 2 types of op-amp based oscillators.
- RC phase shift Oscillator
- Wein Bridge Oscillator.

Wein Bridge Oscillator

In Wein Bridge oscillator, Wein bridge oscillator, it is connected b/w the amp. i/p terminals & o/p terminals. The bridge was a series RC network in one arm & parallel RC

To maintain oscillation, total phase shift across the net must be 0 and loop gain unity. The resonant freq of balanced bridge is given by $f_0 = \frac{1}{2\pi RC}$

Hint for design of Wein Bridge Oscillator with $f_0 = 1\text{kHz}$

1) The gain of sustaining Oscillator (AF) = 3.

2) Oscillating freq $f_0 = \frac{1}{2\pi RC}$

$$AF = 1 + \frac{R_f}{R_i}$$

$$1 + \frac{R_f}{R_i} = 3 \Rightarrow R_f = 2R_i$$

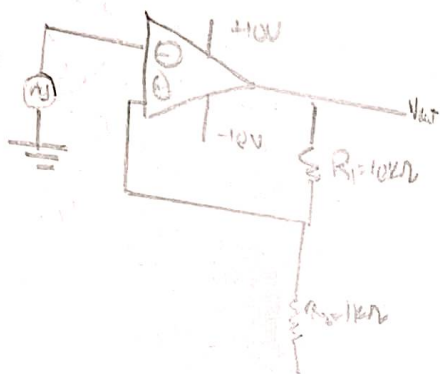
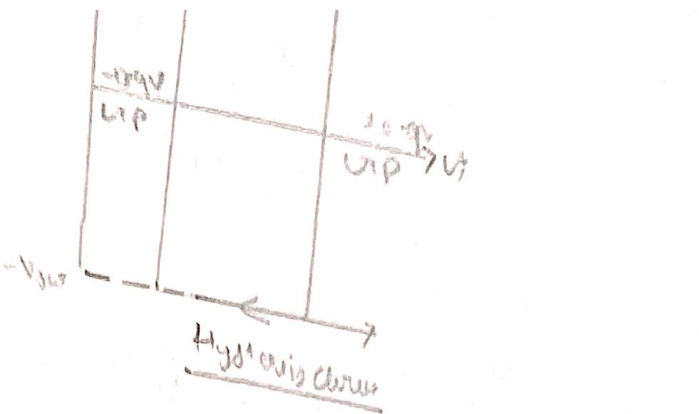
$$\text{Let } R_i = 1\text{K}\Omega \text{ then } R_f = 2\text{K}\Omega$$

$$f_0 = \frac{1}{2\pi RC}$$

$$f_0 = 1\text{kHz}$$

$$C = 0.1\mu\text{F}, R = 1.59\text{K}\Omega$$

Theoretical $F = \frac{1}{\sqrt{3}} \approx 0.577$; Practically, $F = 1/1.7$
The Wein Bridge Oscillator was successfully implemented using op-amp and stable sinusoidal oscillations were obtained. The measured freq & amplitude were found to be close to theoretical values.



From clc.

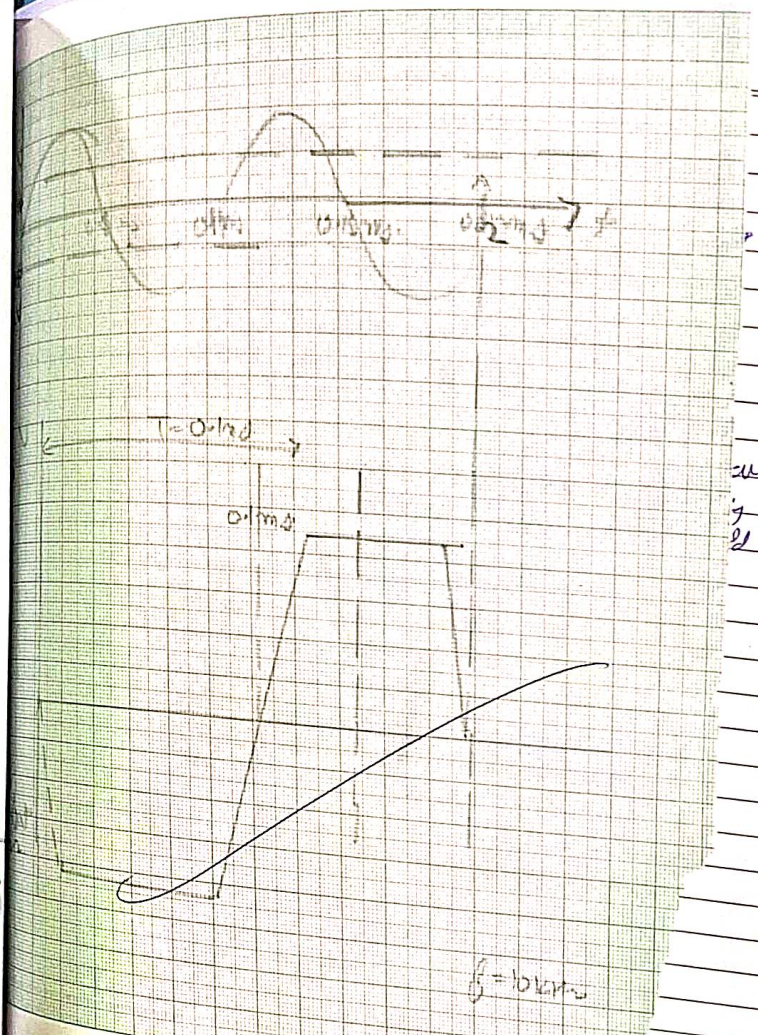
$$V_{out} = B V_{in}$$

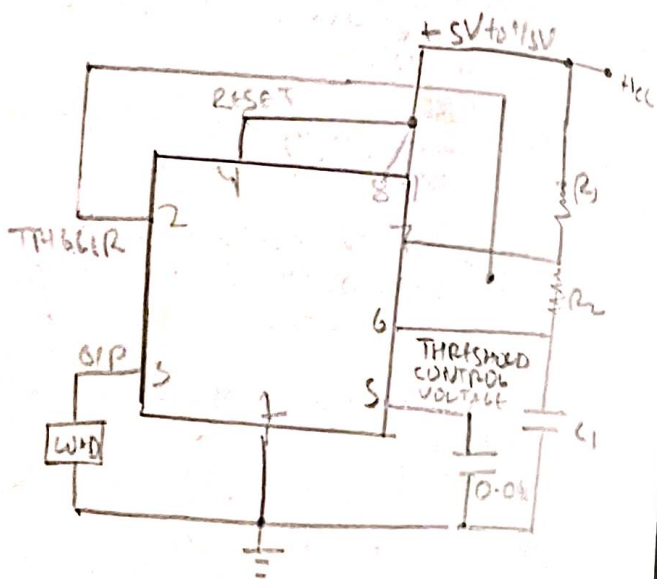
$$V_{out} = 0.00V$$

$$V_{out} = -V_{sat} = -0.00V$$

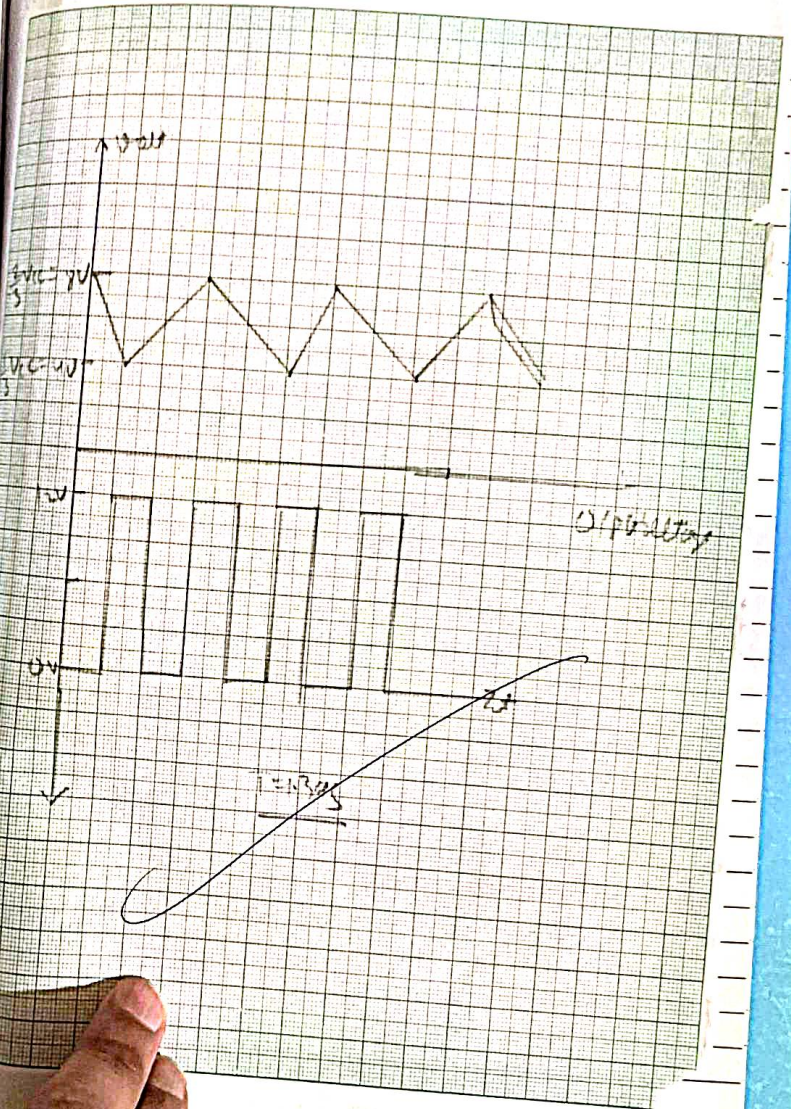
$$Also, B = \frac{R_2}{R_1 + R_2}$$

$$\Rightarrow B = 0.0000$$





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Exp-8

To make and test the operations of Astable Multivibrator using a 555 timer.

Theory

The 555 Timer IC can be connected either in its Monostable mode thereby producing a precise time of a fixed time duration or in its Bistable mode.

The 555 oscillator is another type of abn. relaxation oscillator for generating stabilized square wave or p. waveforms of either fixed freq upto 520 kHz or of varying duty cycle for 50 - 100%.

The capacitor charges through R_1 and R_2 and discharges through R_2 via pin 7. It charges up to $3V_{CC}$ and discharges down to $\frac{1}{3}V_{CC}$.

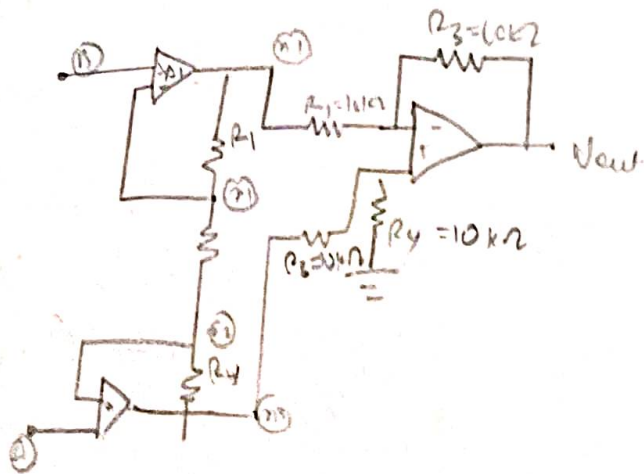
$$t_1 = 0.693 (R_1 + R_2) C \quad (\text{Charging time})$$

$$t_2 = 0.693 R_2 C$$

$$T = t_1 + t_2$$

$$= 0.693 (R_1 + 2R_2) C$$

The 555 Timer IC was successfully configured in astable mode and a continuous square waveform was generated.



1/p Stage

Diffⁿ amplifier

It using Op-amp

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Exp-9

Aim- Design and analysis of an Instrumentation Amplifier for low level signal amplifier.

Def- An Instrumentation Amplifier (IA) is a precise diffⁿ amplifier designed to amplify very low level signal while rejecting noise & interference.

The instr IA typically consists of 3 Op-amps. The 1st stage consists of 2 non-inverting amps.

The 2nd stage is a diffⁿ amp = the difference b/w 2 i/p signal is while rejecting common mode signals.

The gain of an IA is controlled by single resistor making it flexible and easy to set. It is suitable for amplifying very small signals.

$$G = \left(1 + \frac{2R_1}{R_3}\right) \left(\frac{R_2}{R_2}\right)$$

$R_3 =$ Gain setting Resistor.

$R_1, R_2, R_3 =$ Fixed Resistor.

$R_3 = R_2$ then gain simplifies.

$$G = 1 + \frac{2R_1}{R_2}$$

The o/p volt. of the IA is:

$$V_o = G(V_2 - V_1)$$

Result

The IA was successfully implemented and the gain was controlled using an ext. resistor. The op-amp o/p was observed to be an amplified version of i/p signal with good accuracy & high noise rejection.

[Signature]
24/11/26

Exp-10

To study 3rd order Low pass filter.

Op-Amp, Resistor, capacitor, Breadboard, connecting wires.

A LPF is elecⁿ ckt that allows signal with freq lower than a specified cut-off freqⁿ to pass while attenuating signal with freq's higher than cut-off freqⁿ.

The roll off rate of 3rd order LPF is 60dB/decade above f_c .

The overall freqⁿ response of 3rd order LPF exhibits improved selectivity and better attenuation of high frequency signal.

The magnitude of transfer fn of a LPF is-

$$\left| \frac{V_o}{V_{in}} \right| = \frac{A_v}{\sqrt{1 + (f/f_c)^6}}$$

At $\rightarrow$ passband gain of filter.

At 0

Result

The study and graph of LPF has been implemented.